



CSE260 : Digital Logic Design

Group 9 Lab Project Report

Project Name: Majority Voting System**Group Members Information:**

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Introduction:

Digital logic is the core logic system that drives electronic circuit board design. Digital logic is the manipulation of binary data using printed circuit board technology that uses circuits and logic gates to design the execution of computer processes. The first usage of EVM(Electronic Voting Machine) was introduced in the year from 1998 to 2001. We have come a long way from then. Nowadays, every country's main voting system has transitioned from analog to digital. As a result, this technology serves as a foundation for a computerized voting system. A digital circuit that adds two numbers together is known as a complete adder. Full adders are built in hardware using logic gates. And with help of the full adder, 4 bit binary circuit, BCD to 7 segment decoder, and 7 segment common cathode - we have our project to create a majority voting system. The goal of this voting system is to provide a group of individuals with the number of "yes" votes and the number of "no" votes on a topic of interest at the same time. At each assembly point, there is a switch for selecting "yes" or "no," as well as a digital display for the number of yes votes and one for the number of no votes in this project. Furthermore, this approach facilitates voting. Because a large number of people can vote, the majority can be easily ascertained.

Proposed Model:

For a sequential operation to iterate an endless number of times, a fully well balanced circuit model has been constructed. The logic gates and integrated circuits have been constructed in such a way that each gate and flip-flop are interconnected, reducing the need for extra wires and increasing the efficiency of the process.. In this project we took four full adders, 2 ICs (4008 and 4511), 7 segment common cathode display to create a majority voting circuit. The objective of this model is to select the majority among candidates on different occasions easily.

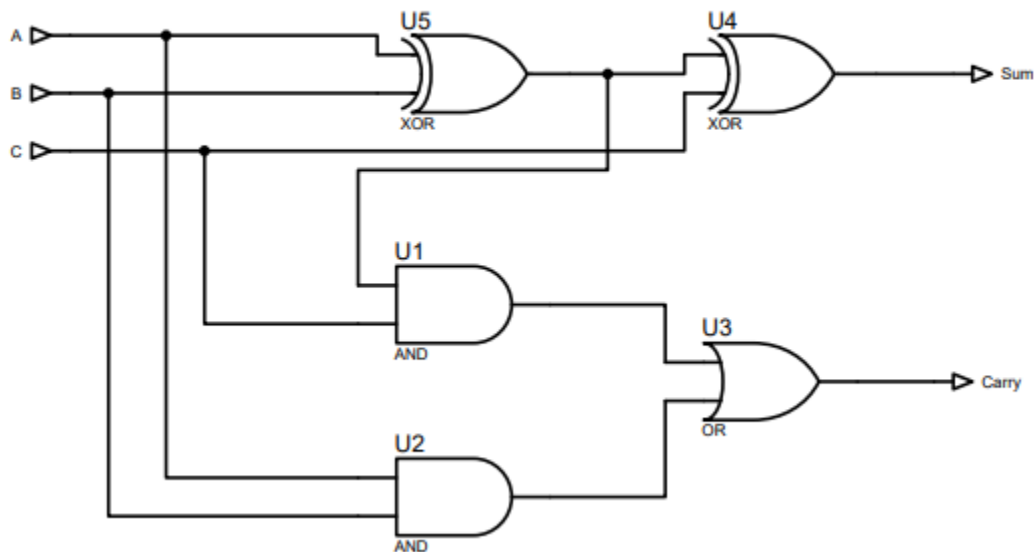
Experimental Setup:

Components Used -

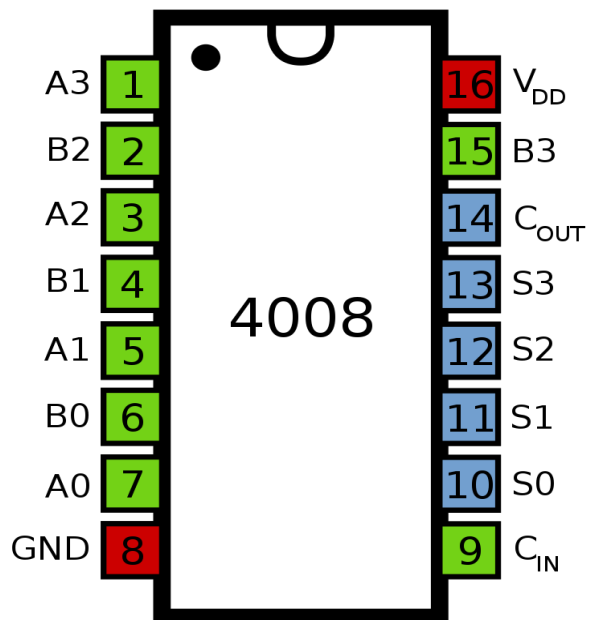
1. Proteus Design Suite Software
2. OR Gates
3. And Gates
4. X-OR Gates
5. LOGIC TOGGLE
6. IC-4008
7. IC4511
8. BCD TO 7- Segment Decoder/LCD Driver
9. Ground

10. 7SEG-COM-CATHODE

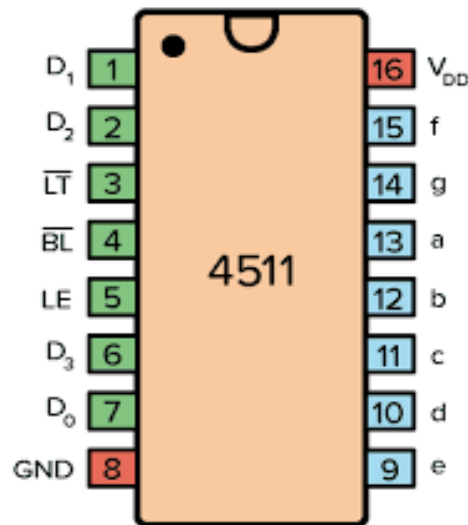
11. Inverter

Step-1`:

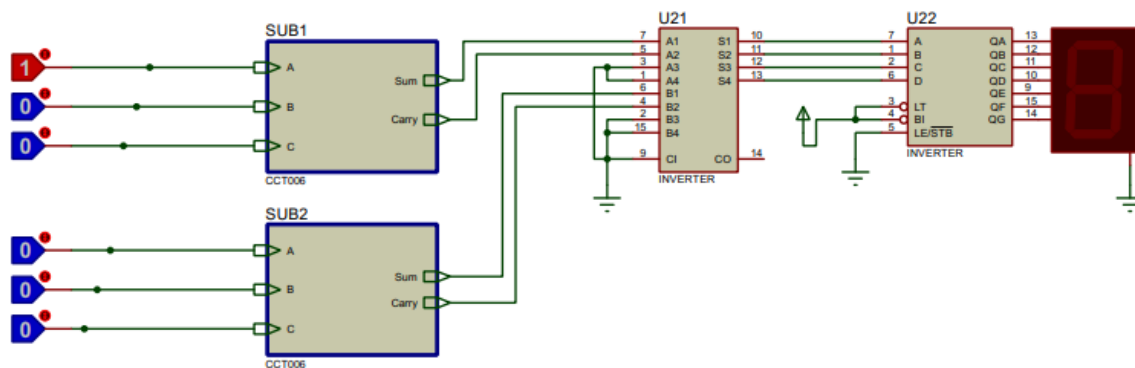
In step-1, we have created a full adder using AND, OR, X-OR gates which takes 3 inputs and gives 2 outputs. A, B and C are the three inputs and alongside among the two outputs one is for sum and another one is for carry.

Step 2:

The CD4008 is a Full Adder with Carry in and Carry out feature. It is a four-stage 4-bit counter, meaning it has four individual Full adder circuits each of 4-bit inside a single package. It can also be easily cascaded if more than four stages are required.

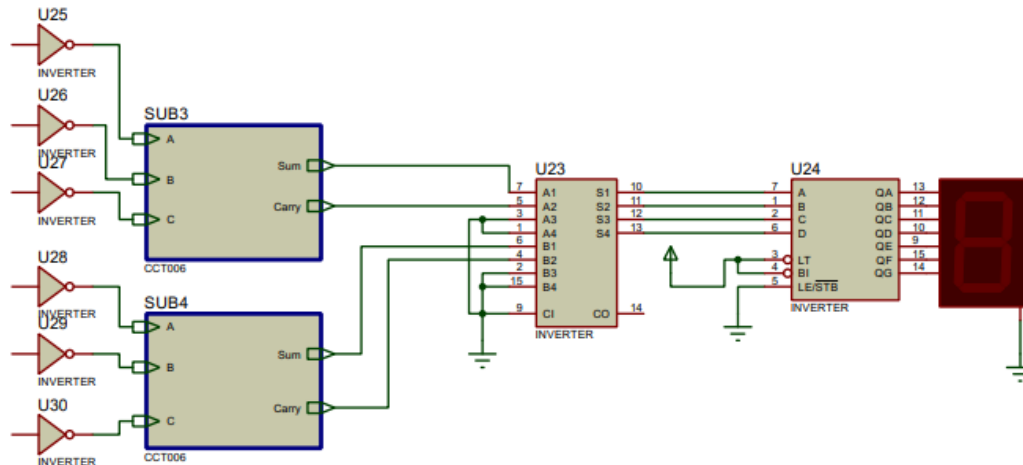


This CD4511 IC is basically a BCD to 7-Segment decoder/driver IC, using which you can convert a binary number into a decimal number on the 7-segment display. We will be able to display a decimal number without using this chipset. IC 4511 has some input pins called BCD. We just have to make those BCD inputs High or Low according to the BCD code of that decimal number and we get the decimal number on display. For example: If you want to display '4', the binary code of four is 0100 so we will give 0,1,0,0 to D, C, B, A inputs respectively, and get the decimal number '4' displayed on 7-segments.



In step-2, we have two full adders, take 3 inputs and give 2 outputs each. Hence, we have 4 outputs of 2 sum and 2 carry. Then, we have placed an IC-4008 where A1 and A2 are connected with output of SUB1 and B1 and B2 are connected with the output of SUB2. Furthermore, we have connected (A3,A4 & B3,B4) with the ground to neutralize them. The 4 outputs we have from IC-4008 are connected as input to a BCD TO 7-Segment Decoder/LCD Driver also with a ground.

Then we've placed a 7-Segment-Common-Cathode to display the final output with red LED.

Step:3

In step 3 we have taken six inverters then connected the first three inverters with the three inputs of SUB3 and the remaining three inverters are connected with the three inputs of SUB4. Afterwards, the rest of the process is the same as we have described in step2. The 4 outputs we have got from IC-4008 are connected as input to a BCD TO 7-Segment Decoder/LCD Driver also with a ground. Finally, IC-4511 gives 7 outputs which are connected to a 7-Segment-Common-Cathode to display. It will receive the binary data and transform it to the digital signal and will put up the final result on the Red LED.

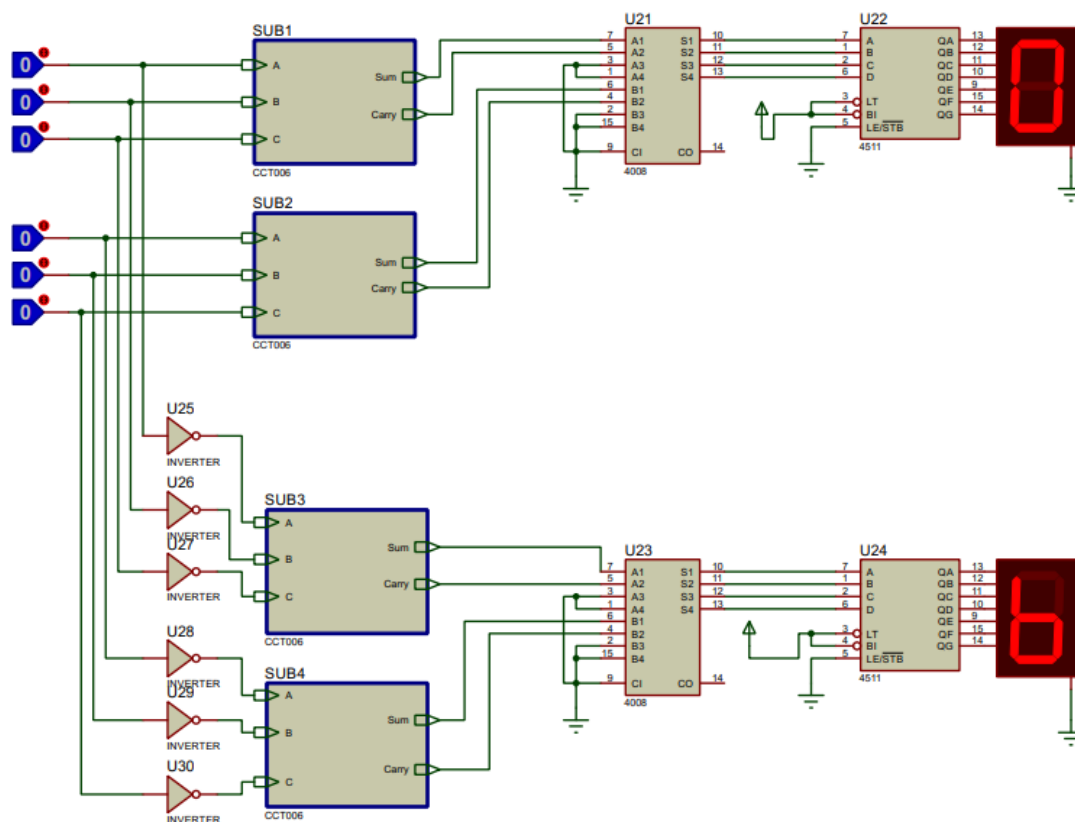
Step:4

Figure of Major voting project

In step 4 we have connected the 6 inverters with the six inputs. We are assuming “yes” as 1 and “no” as 0. Besides if we invert 1, we will get 0. Now the setup is complete and it is working accurately by showing the result in the Red LED.

Results and Analysis:

A full adder takes two binary numbers plus a carry or overflow bit. The output is a sum and another carry bit. Full adders are made from XOR, AND and OR gates in hardware. Full adders are commonly connected to each other to add bits to an arbitrary length of bits, such as 32 or 64 bits. A full adder is effectively two half adders, an XOR and an AND gate, connected by an OR gate.

Truth Table:

Input			Output	
A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Karnaugh Map :**For Sum:**

A \ BC _{in}	BC _{in}			
	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$\text{Sum} = \overline{A}\overline{B}C_{\text{in}} + \overline{A}B\overline{C}_{\text{in}} + A\overline{B}\overline{C}_{\text{in}} + ABC_{\text{in}}$$

For Carry:

A \ BC _{in}	BC _{in}			
	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$C_{out} = AB + A C_{in} + B C_{in}$$

From here on, 2 full adders produce 2 sum and 2 carry. They work as input for IC-4008. The first full adder(Sub1)'s sum & carry are connected to A1,A2, the second ones connected to B1,B2. The outputs are from S1 TO S4. These 4 outputs are connected to BCD to 7 segment decoder as A,B,CD respectively which produces 7 outputs(From QA to QG). Finally, the 7 outputs are connected to a 7 segment common cathode which turns the binary code to digital number. Here, the key is the 4-bit binary output(with fast-carry) we get from IC-4008. This code is turned into 7 segments and then transformed to a digital number.

Conclusion:

Majority voting is a system in which a winning candidate must receive an absolute majority of votes. When a group of people is gathered and it is necessary to quickly determine opinions (for or against), make decisions, or vote on certain subjects or other matters, this type of method can be employed. However, there is one potential flaw in the system. The technique only works for two candidates, which is a disadvantage. The system is unable to determine the majority if there are more than two candidates. If there are three candidates, for example, two of them can be picked and the majority divided between them. The winner will vote alongside the third contender once more. In its most

basic form, however, the system contains a switch for selecting "yes" or "no" at each location in the assembly, as well as a digital display for the number of yes votes and one for the number of negative votes. This circuit serves as the basic foundation for all majority voting systems.